ERA	TIME	TEMP(K)	PARTICLES	FORCES	NOTE
RADIATION	$0-10^{-43}$	OVER 10^{32}	UNKNOWN	(1) UNIFIED	A
	* 10^{-43} s	10^{32}	DARK MATTER?	(2) GRAVITY "FREEZES OUT"	B
	* 10^{-35} s	10^{28}	QUARKS/ELECTRONS	(3) S.F. FREEZES OUT	C
	10^{-10} s	10^{15}	PROTONS/NEUTRONS AND ANTIPARTICALS?	(4) W.F. FREEZES OUT	D
	10^{-4} s	10^{12}	PROTONS/NEUTRONS AND ANTIPARTICALS		E
	2 MINS.-20 MINS.	$10^{9}-10^{7}$	He NUCLEI FORM		F
MATTER	*379,000 YEARS	3000	ATOMS FORM		G
	10^{9} YEARS	100	MILKY WAY FORMS		H
DARK	9.4×10^{9} YEARS	10	SOLAR SYSTEM FORMS		I
ENERGY	1.38×10^{10} YEARS	2.73	TODAY		J

A: PLANCK EPOCH (Q.M.: WE MAY NEVER KNOW) ☹

B: GUT EPOCH (S.N.F., W.N.F., E.M. ARE STILL UNIFIED)

C: QUARK EPOCH. WHEN THE S.N.F. FREEZES OUT, A "FALSE VACUUM" SITUATION IS CREATED. 10^{-35} s → 10^{-32} s: INFLATIONARY EPOCH: GRAVITY IS VERY STRONG AND REPULSIVE. UNIVERSE EXPANDED AS MUCH AS 10^{100} IN SIZE!

D: ALL 4 FORCES ACT AS THEY DO TODAY.

E: PAIR PRODUCTION/ANNIHILATION:

SYMMETRY BREAKING → EVERY 1 BILLION OF THESE, 1 EXTRA PARTICLE IS CREATED ???

F: "PRIMORDIAL NUCLEOSYNTHESIS" → FURIOUS ROUND OF THE P-P CHAIN REACTION ($4H \rightarrow {}^1He$)

G: ATOMIC EPOCH (PERIOD OF DECOUPLING → MATTER AND ENERGY). FIRST PHOTONS OF LIGHT ESCAPE FROM THE PRIMORDIAL MATTER → CMB

H: GALAXY EPOCH: MOST MAJOR GALAXIES ARE FORMING

I: STELLAR EPOCH: 4.6 BILLION YEARS AGO

J: TODAY'S VALUES

IMPORTANT ONES?

3 "BIG" PROBLEMS WITH THE ORIGINAL B.B. THEORY:

① UNIVERSE IS "BUBBLY" (S.C., VOIDS).

C.M.B.R.: 2.73 K IN ALL DIRECTIONS

1980s: COBE (LATER: WMAP) FOUND TINY TEMP. VARIATIONS ($\frac{1}{1{,}000}$ K) IN THE C.M.B.R.

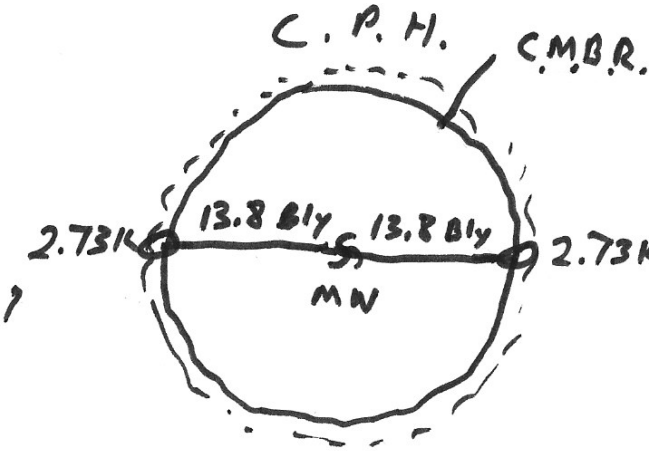
FIRST: ☺ LATER: ☹

1990s: DARK MATTER! NOW: ☺

COOLER → S.C. WARMER → VOIDS

② HORIZON PROBLEM:

HOW DID THESE REGIONS COME TO THE SAME TEMP. (CAN'T COMMUNICATE)??



SOLVED BY INFLATION!

③ FLATNESS PROBLEM:

OUR UNIVERSE MAY BE THE SURFACE ONLY OF A 4 PHYSICAL DIMENSIONAL BALL

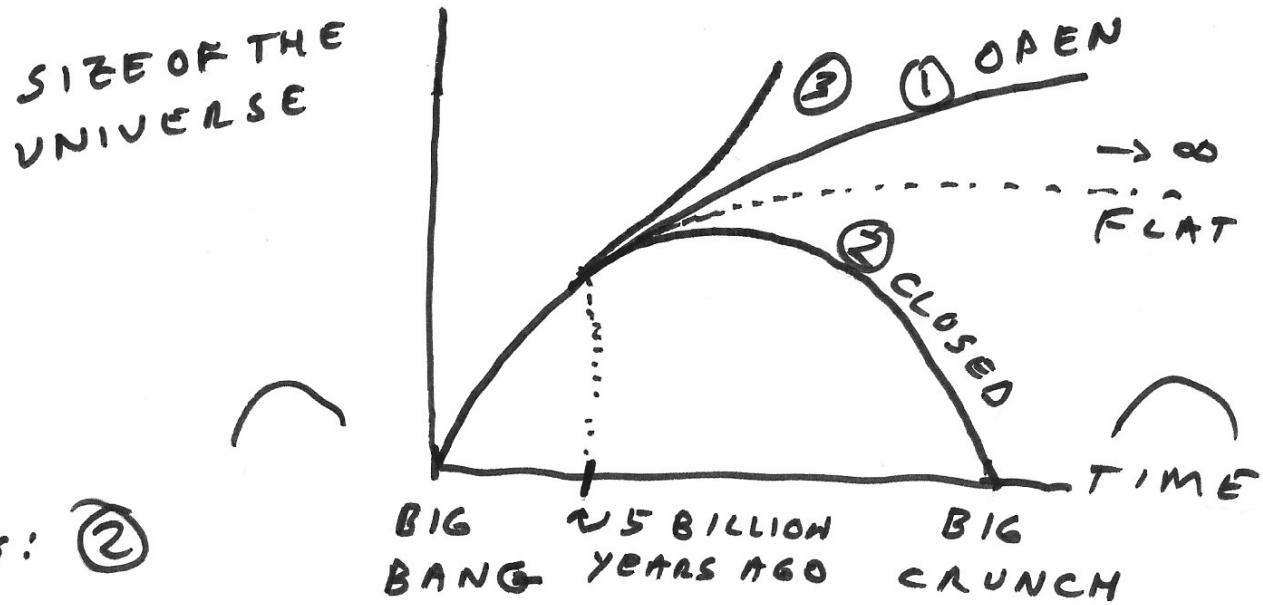
BUT, IT APPEARS FLAT ??

SOLVED BY INFLATION!

TODAY: THE UNIVERSE IS EXPANDING

FUTURE: ① EXPAND FOREVER → ☹

② STOP EXPANDING & RECOLLAPSE → ☺



1970s: ②

1980s: DO WE HAVE THE "CRITICAL DENSITY" (AMOUNT OF MATTER NEEDED TO HALT THE EXPANSION). NOT EVEN CLOSE – ①

1990s: DARK MATTER!

CLOSER ..... BUT STILL ①

H.S.T.: DEEP FIELD (1000S OF GALAXIES)

2000s: IN ONE GALAXY IS A TYPE Ia SUPERNOVA
TOO DIM (TOO DISTANT) FOR ①

THE EXPANSION IS ACCELERATING! ③

EITHER ① OR ③ : EXPAND FOREVER

IN ALL OF SPACE THERE IS "DARK ENERGY"

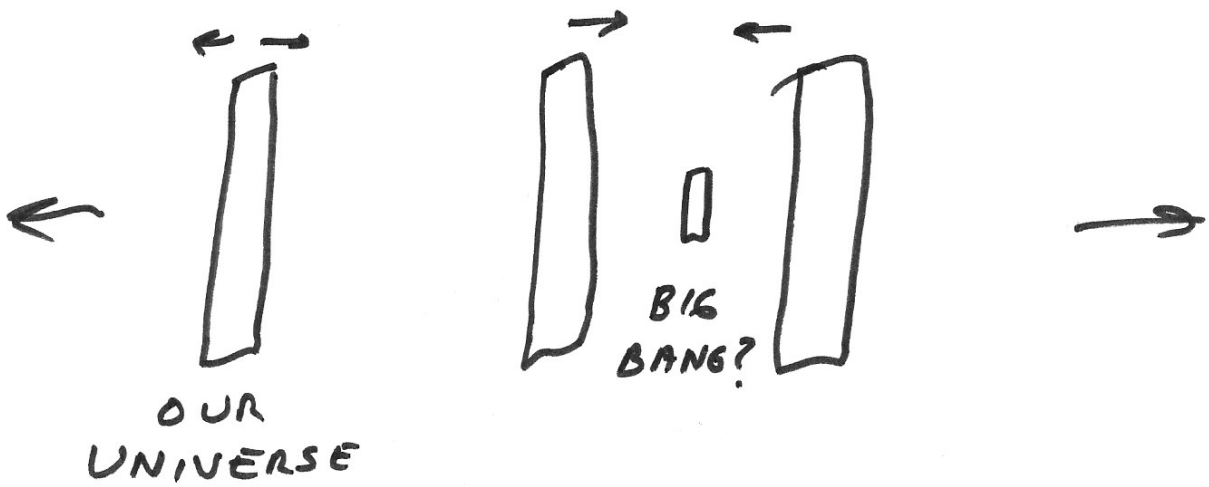
GENERAL RELATIVITY: ADD A C.C.!

$E = mc^2$ UNIVERSE: 73% D.E. (?)
23% D.M. (?)
4% NORMAL MATTER

LOTS MORE TO LEARN!

WHAT CAUSED THE B.B.? POSSIBILITY!

STRING THEORY → M THEORY

OUR UNIVERSE IS 11 DIMENSIONAL!

IT EXISTS IN A 12 DIMENSIONAL "BULK"